

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			80 (2) if answer incorrect award (1) for any of following 84 000 J in method final answer of 0.008 or 8 no ecf possible if formula is rearranged incorrectly or if incorrect energy value taken from table		2		2	2	
	(b)			award (2) for all points plotted correctly - tolerance ± 1 square award (1) for any 3 points plotted correctly award (1) for straight line through all point - ruler must be used		3		3	3	
	(c)			award (2) for high-level quantitative description - as the mass doubles, the energy doubles - mass and energy are directly proportional award (1) for lower-level description - as the mass increases, the energy increases - mass and energy are proportional - mass and energy are directly correlated - mass and energy have a linear relationship		2		2	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			award (1) for resolution and (1) for sensible explanation e.g. change glass beaker to copper can (1) copper is a better conductor (of heat) / increase heat transfer (1) shield the apparatus (1) prevent draughts / reduce heat loss to surroundings (1) lower the beaker (nearer the flame) (1) increase heat transfer / reduce heat loss to surroundings (1) use a lid / insulate the beaker (1) reduce heat loss to surroundings (1)			2	2		2
				Question 8 total	0	7	2	9	7	2